ЦЕНТРАЛЬНЫЙ УНИВЕРСИТЕТ

1 multiply-accumulate = 2 FLOPs

FLOPs(S,B) =

for conv 7x7 s2 3→32

= 2·B·S/2·S/2·32·7·7 +

for conv 5x5 32→64

+ 2·B·S/4·S/4·64·32·5·5 +

for conv 3x3 s2 64→128

+ 2·B·S/8·S/8·128·64·3·3 +

for conv 1x1 128→256

+ 2·B·S/8·S/8·256·128·1·1 +

for conv 3x3 s2 256→256

+ 2·B·S/16·S/16·256·256·3·3 +

for conv 1x1 256→512

+ 2·B·S/16·S/16·512·256·1·1 +

for Linear 512→256

+ 2·B·512·256 +

for Linear 256→100

+ 2·B·256·100

$Memory(S,B) = M_{weights} + \max(M_{input} + M_{output} + M_{workspace})$. Every parameter is 4 bytes since FP32

$M_{weights}$ =

for conv 7×7, 3→32

$4 \cdot 32 \cdot 3 \cdot 7 \cdot 7 +$

for conv 5×5, 32→64

$+ 4 \cdot 64 \cdot 32 \cdot 5 \cdot 5 +$

for conv 3×3, 64→128

$+ 4 \cdot 128 \cdot 64 \cdot 3 \cdot 3 +$

for conv 1×1, 128→256

$+ 4 \cdot 256 \cdot 128 \cdot 1 \cdot 1 +$

for conv 3×3, 256→256

$+ 4 \cdot 256 \cdot 256 \cdot 3 \cdot 3 +$

for conv 1×1, 256→512

$+ 4 \cdot 512 \cdot 256 \cdot 1 \cdot 1 +$

for Linear 512→256

$+ 4 \cdot 512 \cdot 256 +$

for Linear 256→100 $+ 4 \cdot 256 \cdot 100$

without biases

bytes

ЦЕНТРАЛЬНЫЙ УНИВЕРСИТЕТ

For activation peak:

$M_{input} = 4 \cdot B \cdot 3 \cdot S \cdot S$

$= 4 \cdot B \cdot 32 \cdot S/2 \cdot S/2$

$M_{conv7x7,s2,3\to32}$

- $M_{peak_1} = M_{input} + M_{conv1x1,s2,3\to32}$

$M_{beforepool} = 4 \cdot B \cdot 32 \cdot S/2 \cdot S/2$

$M_{afterpool} = 4 \cdot B \cdot 32 \cdot S/4 \cdot S/4$

- $M_{pool} = M_{beforepool} + M_{afterpool}$

$M_{input_2} = 4 \cdot B \cdot 32 \cdot S/4 \cdot S/4$

$M_{conv5x5,32\to64} = 4 \cdot B \cdot 64 \cdot S/4 \cdot S/4$

- $M_{peak_2} = M_{input_2} + M_{conv5x5,32\to64}$

$M_{input_3} = 4 \cdot B \cdot 64 \cdot S/4 \cdot S/4$

$M_{conv3x3,s2,64\to128} = 4 \cdot B \cdot 128 \cdot S/8 \cdot S/8$

- $M_{peak_3} = M_{input_3} + M_{conv3x3,s2,64\to128}$

$M_{input4} = 4 \cdot B \cdot 128 \cdot S/8 \cdot S/8$

$M_{conv\,1\times1,\,128\to256} = 4 \cdot B \cdot 256 \cdot S/8 \cdot S/8$

- $M_{peak4} = M_{input4} + M_{conv\,1\times1,\,128\to256}$

$M_{input5} = 4 \cdot B \cdot 256 \cdot S/8 \cdot S/8$

$M_{conv\,3\times3,\,s2,\,256\to256} = 4B \cdot 256 \cdot S/16 \cdot S/16$

- $M_{peak5} = M_{input5} + M_{conv\,3\times3,\,s2,\,256\to256}$

$M_{input6} = 4 \cdot B \cdot 256 \cdot S/16 \cdot S/16$

$M_{conv\,1\times1,\,256\to512} = 4 \cdot B \cdot 512 \cdot S/16 \cdot S/16$

- $M_{peak6} = M_{input6} + M_{conv\,1\times1,\,256\to512}$

$M_{input7} = 4 \cdot B \cdot 512 \cdot S/16 \cdot S/16$

$M_{GAP} = 4 \cdot B \cdot 512$

- $M_{peak7} = M_{input7} + M_{GAP}$

$M_{input8} = 4 \cdot B \cdot 512$

$M_{Linear\,512\to256} = 4 \cdot B \cdot 256$

- $M_{peak8} = M_{input8} \cdot M_{Linear\,512\to256}$

$M_{input9} = 4 \cdot B \cdot 256$

$M_{Linear\,256\to100} = 4 \cdot B \cdot 100$

- $M_{peak9} = M_{input9} + M_{Linear\,256\to100}$

ЦЕНТРАЛЬНЫЙ УНИВЕРСИТЕТ

$$Memory(S,B) = M_{weights} + \max(M_{peak_1}, \dots, M_{peak_9})$$

$$Latency(S,B,\Theta) = T_0 + \max\left(\frac{FLOPs(S,B)}{P}, \frac{Bytes_m(S,B)}{BW}\right)$$

$$Bytes(S,B) = \Sigma_{input\,read} + \Sigma_{weight} + \Sigma_{activations} =$$

$$= 4\cdot B\cdot 3\cdot 8\cdot S + 4\cdot 32\cdot 5\cdot 7\cdot 7 +$$
$$+ 4\cdot B\cdot 32\cdot 5/2\cdot 5/2 +$$
$$+ 32\cdot B\cdot 5\cdot 5 + 32\cdot B\cdot 5\cdot 5 +$$
$$+ 4\cdot B\cdot 32\cdot 5/2\cdot 5/2 + 4\cdot B\cdot 32\cdot 5/4\cdot 5/4 +$$
$$+ 8\cdot B\cdot 5\cdot 5 + 4\cdot 64\cdot 32\cdot 25 + 16\cdot B\cdot 5\cdot 5$$
$$+ 16\cdot B\cdot 5\cdot 5 + 16\cdot B\cdot 5\cdot 5 +$$
$$4\cdot B\cdot 64\cdot 5/4\cdot 5/4 + 128\cdot 64\cdot 3\cdot 3 +$$
$$+ 4\cdot B\cdot 128\cdot 5/8\cdot 5/8 +$$

$$+ 4\cdot B\cdot 12$$
$$+ 4\cdot B\cdot 12$$
$$+ 256\cdot 12$$
$$+ 4\cdot B\cdot 2$$
$$+ 16\,B\cdot$$
$$+ 4\cdot B\cdot$$
$$+ 4\cdot B\cdot 25$$
$$+ 4\cdot B\cdot 5\cdot 5$$
$$+ 4\cdot B\cdot 25$$
$$+ 4\cdot B\cdot 5$$
$$+ 8\,B\,S^2$$
$$+ 8\,B\,S^2 +$$
$$+ 4\,B\cdot 5$$
$$+ 1024\,B$$
$$+ 4\cdot 3\cdot$$

$$Energy(S,$$
$$+$$

$$
\begin{aligned}
&+ 4\cdot B\cdot 128\cdot S/8\cdot S/8 + \\
&+ 4\cdot B\cdot 128\cdot S/8\cdot S/8 + \\
&+ 256\cdot 128\cdot T + \\
&+ 4\cdot B\cdot 256\cdot S/8\cdot S/8 + \\
&+ 16B\cdot S\cdot S + 16\cdot B\cdot S\cdot S + \\
&+ 4\cdot B\cdot 256\cdot S/8\cdot S/8 + 4\cdot 256\cdot 256\cdot 9 + \\
&+ 4\cdot B\cdot 256\cdot S/16\cdot S/16 + \\
&+ 4\cdot B\cdot S\cdot S + 4\cdot B\cdot S\cdot S + \\
&+ 4\cdot B\cdot 256\cdot S/16\cdot S/16 + 4\cdot 512\cdot 256 + \\
&+ 4\cdot B\cdot 512\cdot S^2/256 + \\
&+ 8BS^2 + 8BS^2 + \\
&+ 8BS^2 + 2048B + \\
&+ 4B\cdot 512 + 4\cdot 512\cdot 256 + 4\cdot B\cdot 256 + \\
&+ 1024B + 1024B + \\
&+ 4\cdot 3\cdot 256 + 4\cdot 456\cdot 100 + 4B\cdot 100
\end{aligned}
$$

$$
\text{Energy}(S,B,\theta_e) = E_0 + d\cdot \text{Floors}(S,B) + \beta\cdot \text{Bytes}(S,B)
$$